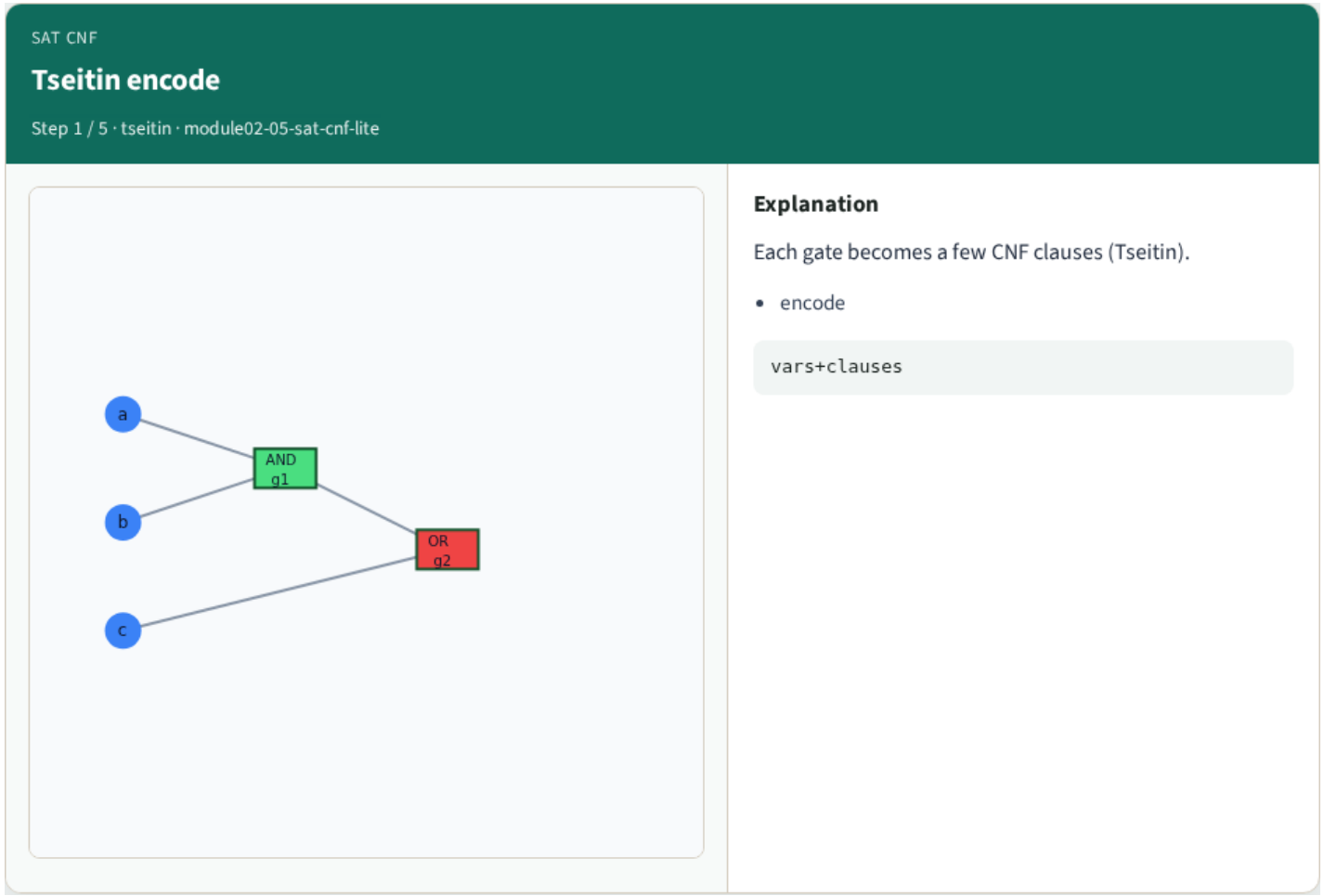
SAT CNF lite

Industrial CEC uses SAT on a miter , first learn clause count on a tiny netlist

The idea

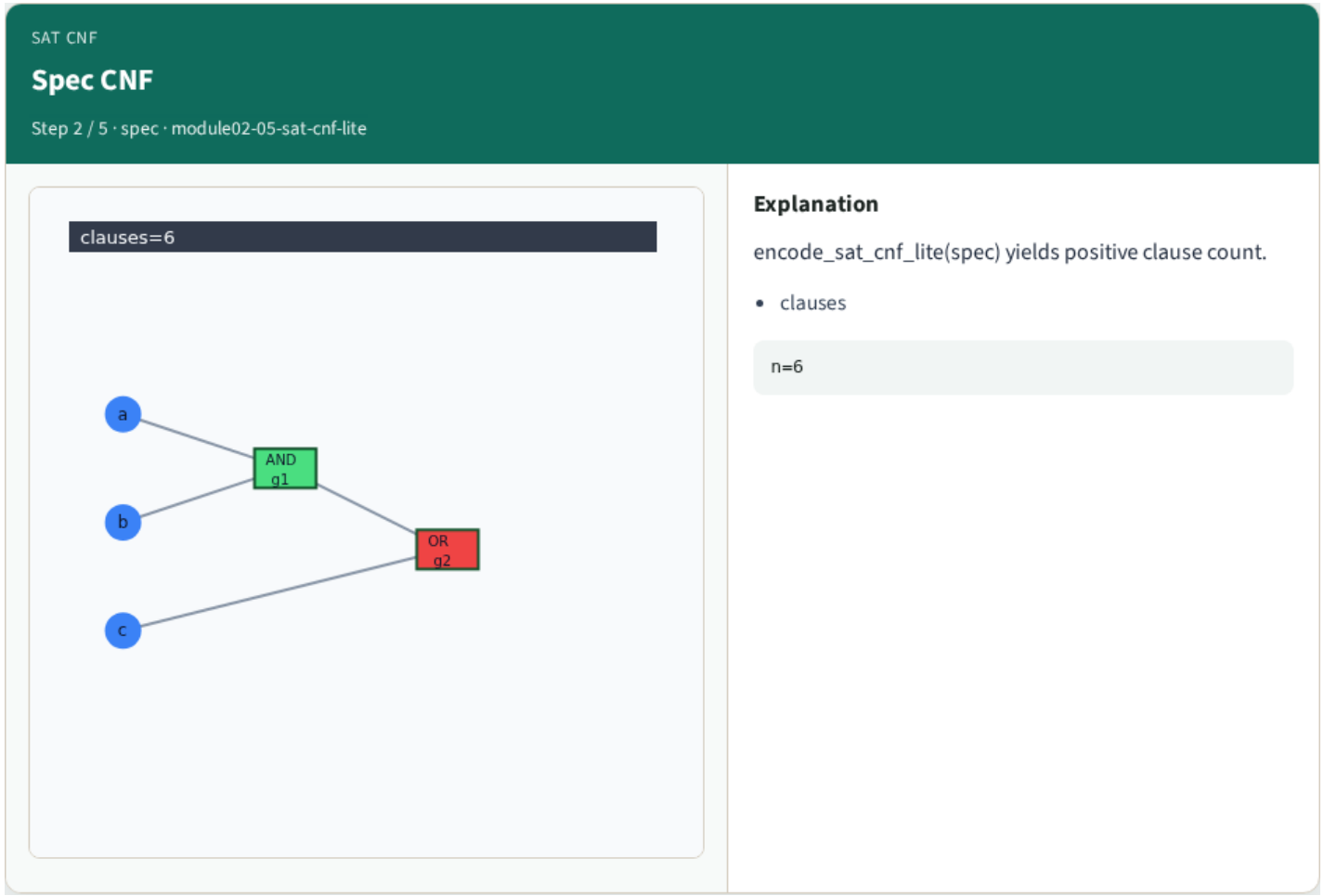
- Encode_sat_cnf_lite assigns PI vars 1..3 and adds clauses per AND/OR/NOT gate

Tseitin encode



Tseitin encode

Spec CNF



Spec CNF

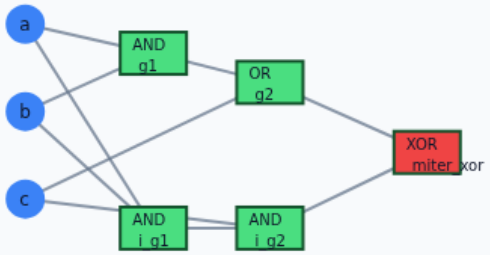
Assert miter=1

SAT CNF

Assert miter=1

Step 3 / 5 · assert · module02-05-sat-cnf-lite

clauses=17



Explanation

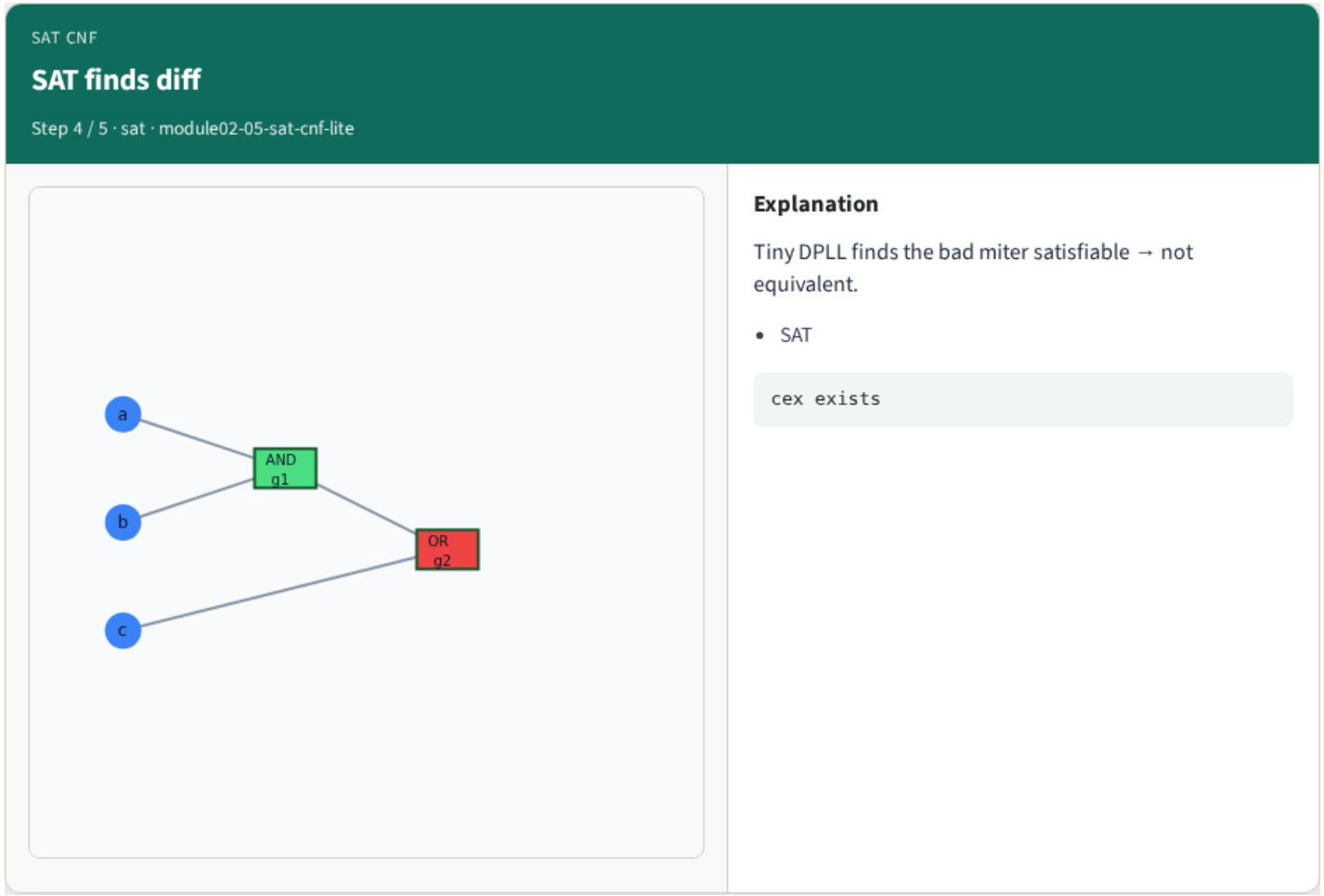
Ask SAT: can bad miter output be 1?

- SAT query

```
diff?
```

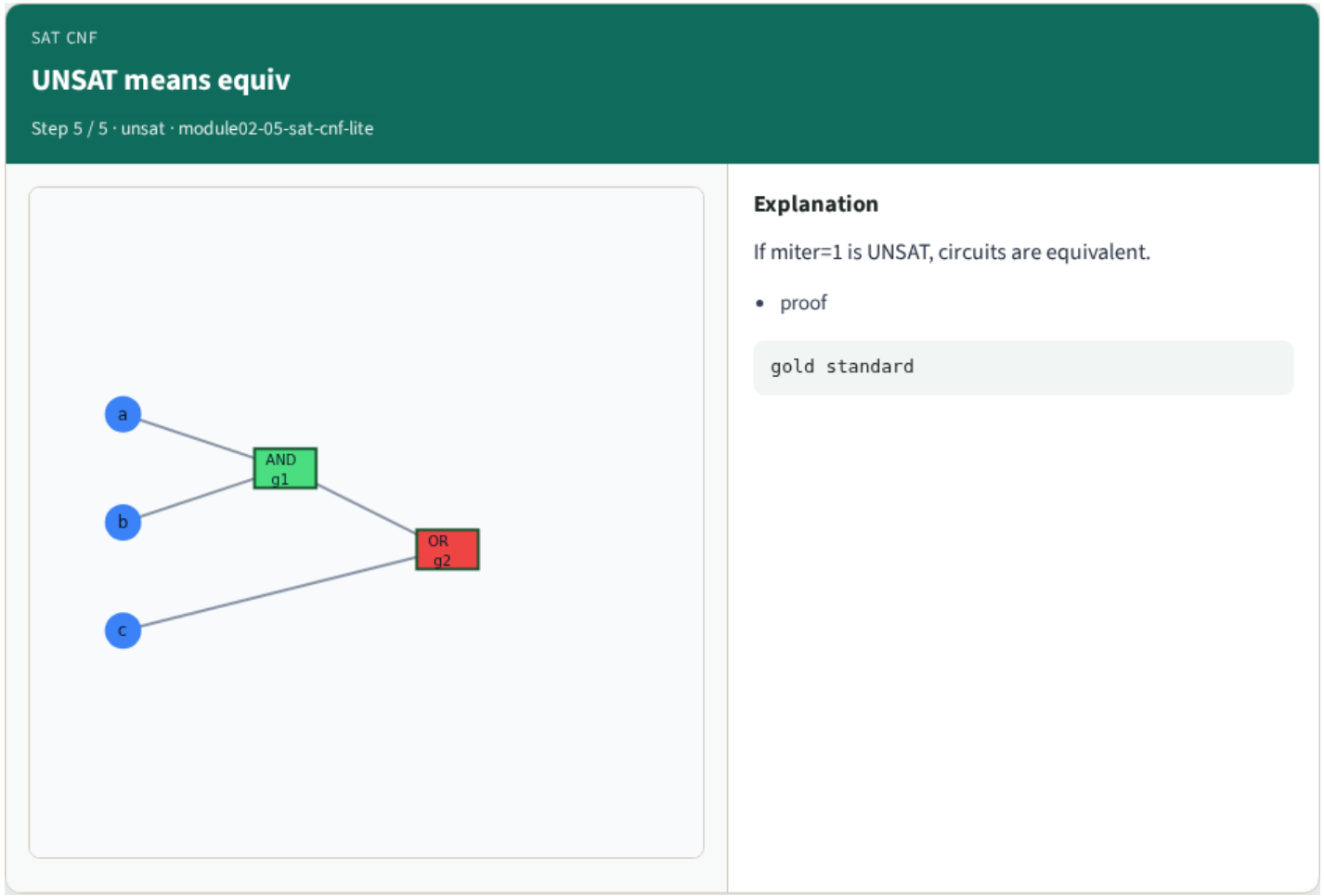
Assert miter=1

SAT finds diff



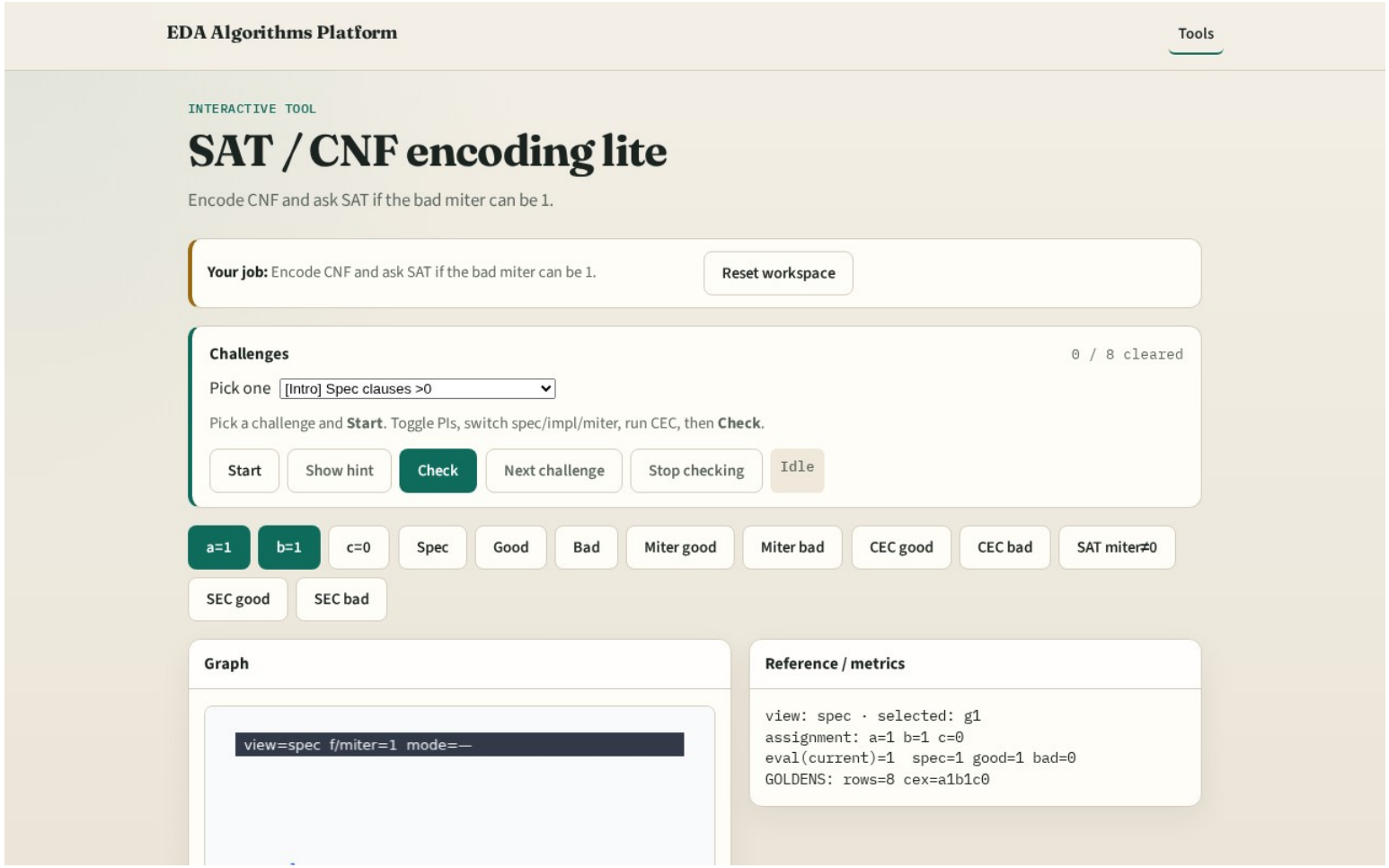
SAT finds diff

UNSAT means equiv



UNSAT means equiv

Browser lab track



Browser lab starter

Implement track

- Call encode_sat_cnf_lite(spec)

Pitfalls

- CNF on outputs without Tseitin auxiliaries
- Expecting full Glucose run

Your turn

- Next: counterexample debug from CEC